

Agent task: max-pack concurrency — Vera vs Zen 5 (9575)

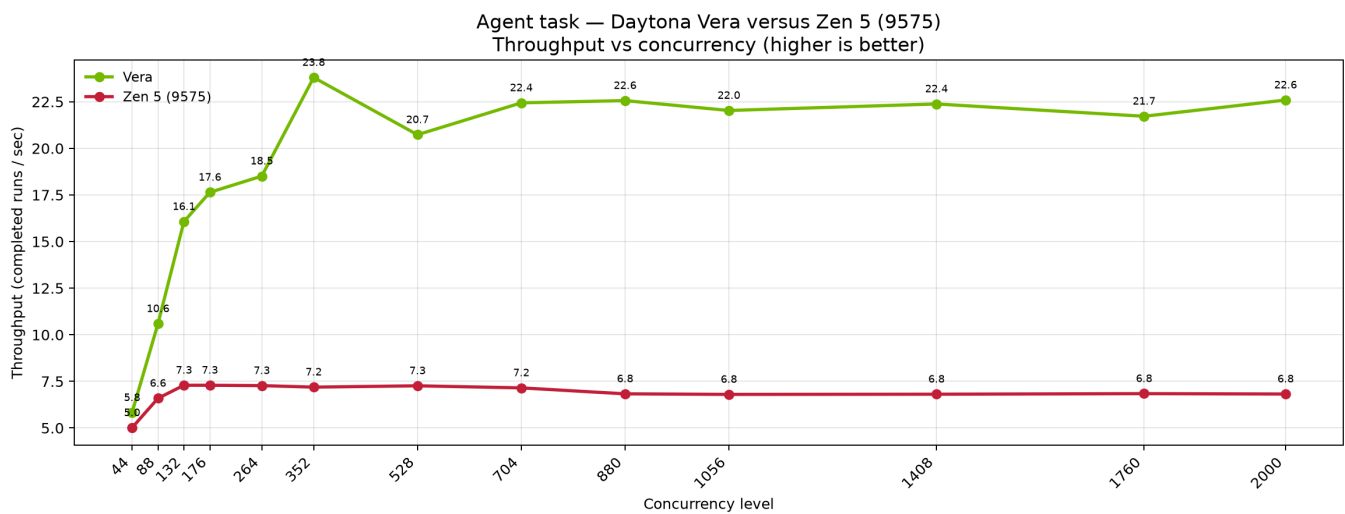
August 2026. Same workload, same seed — matched concurrency runs **44** through **2,000**.

Charts compare **Vera** and **Zen 5 (9575)** max-pack runs (matched ladder through **2,000**).

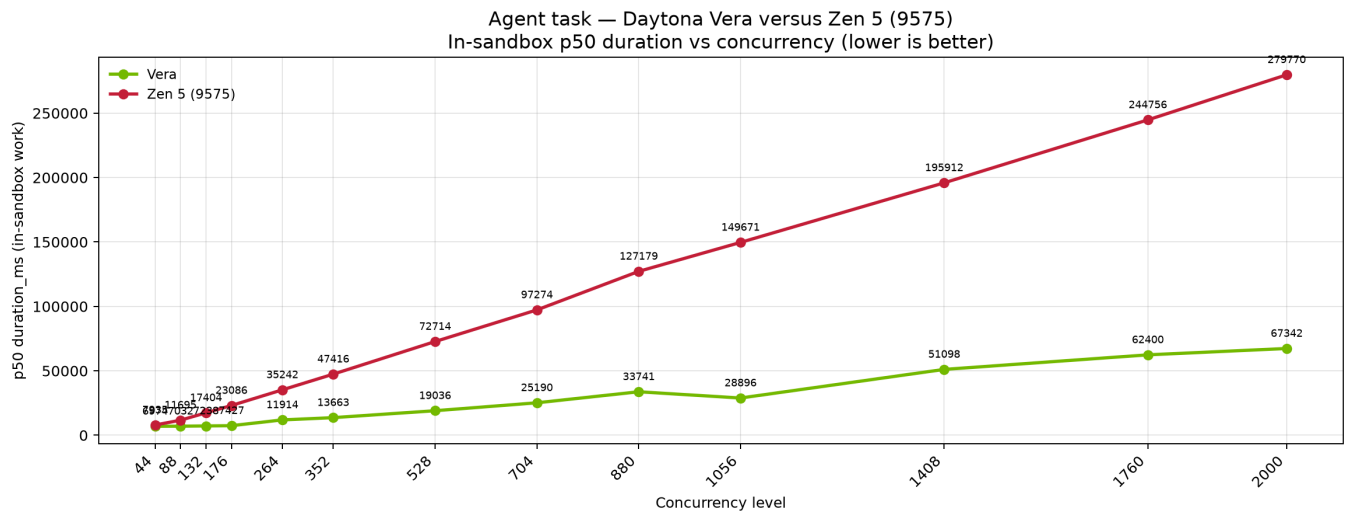
What to take away

On **Vera**, throughput stays near **~22 jobs/s** from 880 through 2,000. On **Zen 5 (9575)**, throughput stays flat near **~6.8 jobs/s** over the same concurrency range — about **3×** lower aggregate packing on the smaller 128-thread box.

Both series use the **same concurrency ladder** on the x-axis through **2,000**.



Throughput = completed runs / exec wall. Max-pack guarantees differ; burst caps match (1 vCPU, 4 GB RAM).



p50 duration_ms is in-sandbox work only (excludes create/delete and client RTT). Max-pack guarantees differ; burst caps match (1 vCPU, 4 GB RAM).

How to read the charts

Throughput (jobs per second) is completed sandbox episodes divided by the wave's exec wall clock. Higher is better.

p50 duration_ms is the median time spent *inside* the sandbox doing the agent loop. Lower is better.

Fair comparison (why this is as matched as we can make it)

Both cells ran the **same agent task** (repo-agent-v3), **same snapshot** (dtgraviet/vera-agent-benchmark:v3), **same** `--n 50`, **seed 42**, `-E 8`, and `--hold-then-exec`. Both use the **identical concurrency ladder** — every rung from **44** through **2,000**:

44 88 132 176 264 352 528 704 880 1056 1408 1760 2000

At concurrency 44 through 704, both cells used the **same reserved capacity**: 0.125 vCPU and 1 GiB memory per sandbox. Burst caps are the same on both (1 vCPU, 4 GB RAM).

From 880 onward, memory and CPU **guarantees** are smaller so each cell can pack **2,000** live sandboxes. **Burst caps are unchanged** on both:

	Vera	Zen 5 (9575)
CPU guarantee	0.125 vCPU	0.025 vCPU
CPU burst cap	1 vCPU	1 vCPU
Memory guarantee	512 MiB	100 MiB
Memory burst cap	4 GB	4 GB

The 0.025 vCPU / 100 MiB guarantees on Zen 5 (9575) do not throttle in-sandbox performance. They are **reservation / packing knobs** on the runner ledger so more sandboxes can stay live on a 128-thread box. Each sandbox still **bursts to a full 1 vCPU and up to 4 GB RAM** during the agent episode — the same burst ceiling as Vera (RLP_BURST_MAX_CPU=1, RLP_BURST_MAX_MEM_MIB=4096 on the cell). We needed the smaller guarantees to pack **2,000** concurrent tenants and match Vera's concurrency axis; with 0.125 vCPU the cell tops out around **~1,013** sandboxes regardless of RLP_MAX_LIVE_VMS.

Configuration

Setting	Reserved capacity (c=44..704)	Vera max-pack (c=880+)	Zen 5 (9575) max-pack (c=880+)
Target	vera / zen5-9575	<code>--target vera</code>	<code>--target zen5-9575</code>
CPU guarantee	0.125 vCPU	0.125 vCPU	0.025 vCPU
CPU burst cap	1.0 vCPU	1.0 vCPU	1.0 vCPU
Memory guarantee	1 GiB (default)	512 MiB	100 MiB
Memory burst cap	4 GB	4 GB	4 GB
Shared	<code>--n 50 --seed 42 -E 8 --hold-then-exec · dtgraviet/vera-agent-benchmark:v3</code>	same	same

Vera — two runs, matched ladder

Step 1 — reserved capacity (1 GiB, c=44..704):

```
UV_NO_SYNC=1 uv run main.py --benchmark agent --runner rlp --target vera \
--snapshot dtgraviet/vera-agent-benchmark:v3 \
--levels 44 88 132 176 264 352 528 704 \
--n 50 --seed 42 -E 8 --hold-then-exec \
--rlp-cpu 0.125 --rlp-cpu-max 1
```

Step 2 — max-pack (512 MiB guarantee, c=880..2000):

```
UV_NO_SYNC=1 uv run main.py --benchmark agent --runner rlp --target vera \
--snapshot dtgraviet/vera-agent-benchmark:v3 \
--levels 880 1056 1408 1760 2000 \
--n 50 --seed 42 -E 8 --hold-then-exec \
--rlp-cpu 0.125 --rlp-cpu-max 1 --rlp-memory 0.5 --rlp-memory-max 4
```

Zen 5 (9575) — two runs, matched ladder

Step 1 — reserved capacity (1 GiB, c=44..704; same levels as Vera):

```
UV_NO_SYNC=1 uv run main.py --benchmark agent --runner rlp --target zen5-9575 \
--snapshot dtgraviet/vera-agent-benchmark:v3 \
--levels 44 88 132 176 264 352 528 704 \
--n 50 --seed 42 -E 8 --hold-then-exec \
--rlp-cpu 0.125 --rlp-cpu-max 1
```

Step 2 — max-pack (100 MiB guarantee, c=880..2000):

```
UV_NO_SYNC=1 uv run main.py --benchmark agent --runner rlp --target zen5-9575 \
--snapshot dtgraviet/vera-agent-benchmark:v3 \
--levels 880 1056 1408 1760 2000 \
--n 50 --seed 42 -E 8 --hold-then-exec \
--rlp-cpu 0.025 --rlp-cpu-max 1 --rlp-memory 0.1 --rlp-memory-max 4 --rlp-disk 1
```

Runbook (Zen 5 (9575) cell prep): [../tickets/redswitches-2k-maxpack-run.md](https://github.com/dtgraviet/vera-agent-benchmark/blob/main/tickets/redswitches-2k-maxpack-run.md)

Headline numbers (matched rungs)

Through 704 = same reserved capacity on both. 880–2,000 = max-pack.

Concurrency	Vera p50 duration	Zen 5 (9575) p50 duration	Vera throughput	Zen 5 (9575) throughput
704	25,190 ms	97,274 ms	22.44 /s	7.15 /s
880	33,741 ms	127,179 ms	22.56 /s	6.83 /s
1,056	28,896 ms	149,671 ms	22.03 /s	6.80 /s
1,408	51,098 ms	195,912 ms	22.38 /s	6.81 /s
1,760	62,400 ms	244,756 ms	21.72 /s	6.84 /s
2,000	67,342 ms	279,770 ms	22.59 /s	6.82 /s

Method notes

- Charts merge Step 1 + Step 2 JSONL per series. Regenerate: `uv run python scripts/nvidia_brief_maxpack_charts.py`
- Base 704 narrative: [../nvidia-agent-brief-704-zen5/](#)
- Sources: [sources.md](#) · Inventory: [maxpack-data-inventory.md](#)